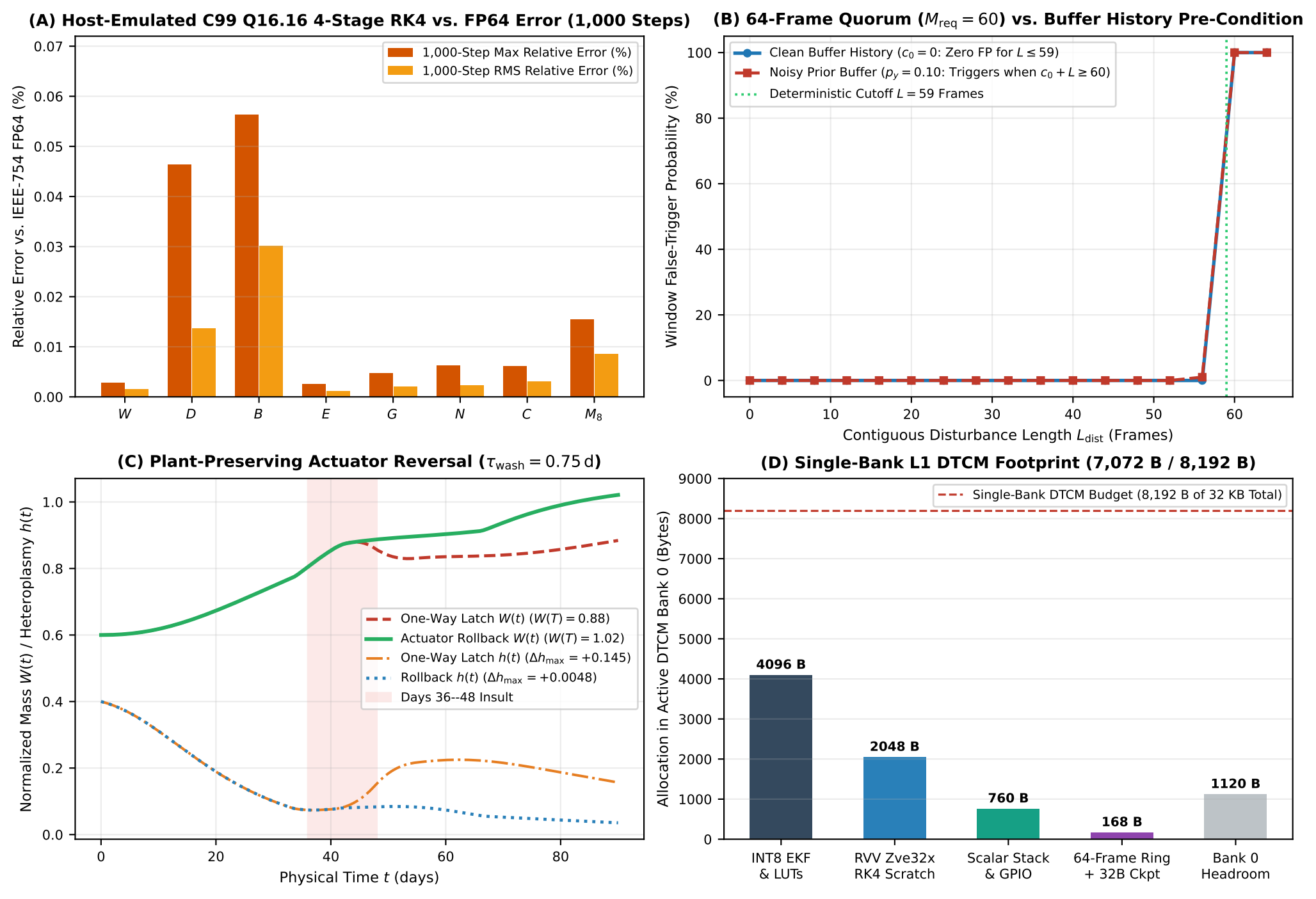

(A) Host-Emulated C99 Q16.16 4-Stage RK4 vs. FP64 Error (1,000 Steps)
1,000-Step Max Relative Error (%)
1,000-Step RMS Relative Error (%)
Relative Error vs. IEEE-754 FP64 (%)
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
W
D
B
E
G
N
C
M8
(B) 64-Frame Quorum ($M_{req} = 60$) vs. Buffer History Pre-Condition
Clean Buffer History ($c_0 = 0$: Zero FP for $L \le 59$)
Noisy Prior Buffer ($p_y = 0.10$: Triggers when $c_0 + L \ge 60$)
Deterministic Cutoff $L = 59$ Frames
Window False-Trigger Probability (%)
0
20
40
60
80
100
0
10
20
30
40
50
60
Contiguous Disturbance Length L_{dist} (Frames)
(C) Plant-Preserving Actuator Reversal ($\tau_{wash} = 0.75$ d)
One-Way Latch $W(t)$ ($W(T) = 0.88$)
Actuator Rollback $W(t)$ ($W(T) = 1.02$)
One-Way Latch $h(t)$ ($\Delta h_{max} = +0.145$)
Rollback $h(t)$ ($\Delta h_{max} = +0.0048$)
Days 36--48 Insult
Normalized Mass $W(t)$ / Heteroplasmy $h(t)$
0.0
0.2
0.4
0.6
0.8
1.0
0
20
40
60
80
Physical Time t (days)
(D) Single-Bank L1 DTCM Footprint (7,072 B / 8,192 B)
Single-Bank DTCM Budget (8,192 B of 32 KB Total)
Allocation in Active DTCM Bank 0 (Bytes)
0
1000
2000
3000
4000
5000
6000
7000
8000
9000
4096 B
2048 B
760 B
168 B
1120 B
INT8 EKF & LUTs
RVV Zve32x RK4 Scratch
Scalar Stack & GPIO
64-Frame Ring + 32B Ckpt
Bank 0 Headroom